

Day 1: LoRA vs DoRA on OpenVLA-7B (vla-lab benchmark)

Concept: how each method updates W

Full Fine-Tune: $W_{\text{new}} = W_{\text{pretrain}} + \Delta W$ (all 7B params trainable)

LoRA: $W_{\text{new}} = W_{\text{pretrain}} + \text{scale} \cdot BA$ (B,A: low-rank, $\sim 30M$)

DoRA: $W_{\text{new}} = m \times (W_{\text{pretrain}} + \text{scale} \cdot BA) / \|m\|_c$
↑ magnitude (one scalar per output)

DoRA = LoRA on direction + dedicated trainable magnitude vector

